

HLD Case Study 8: Real-Time Matching & Recommendation (Tinder)

Key Focus Areas: Bidirectional mutual matching, real-time swipe ingestion, two-stage recommendation ranking, geospatial bucketing, and match notifications.

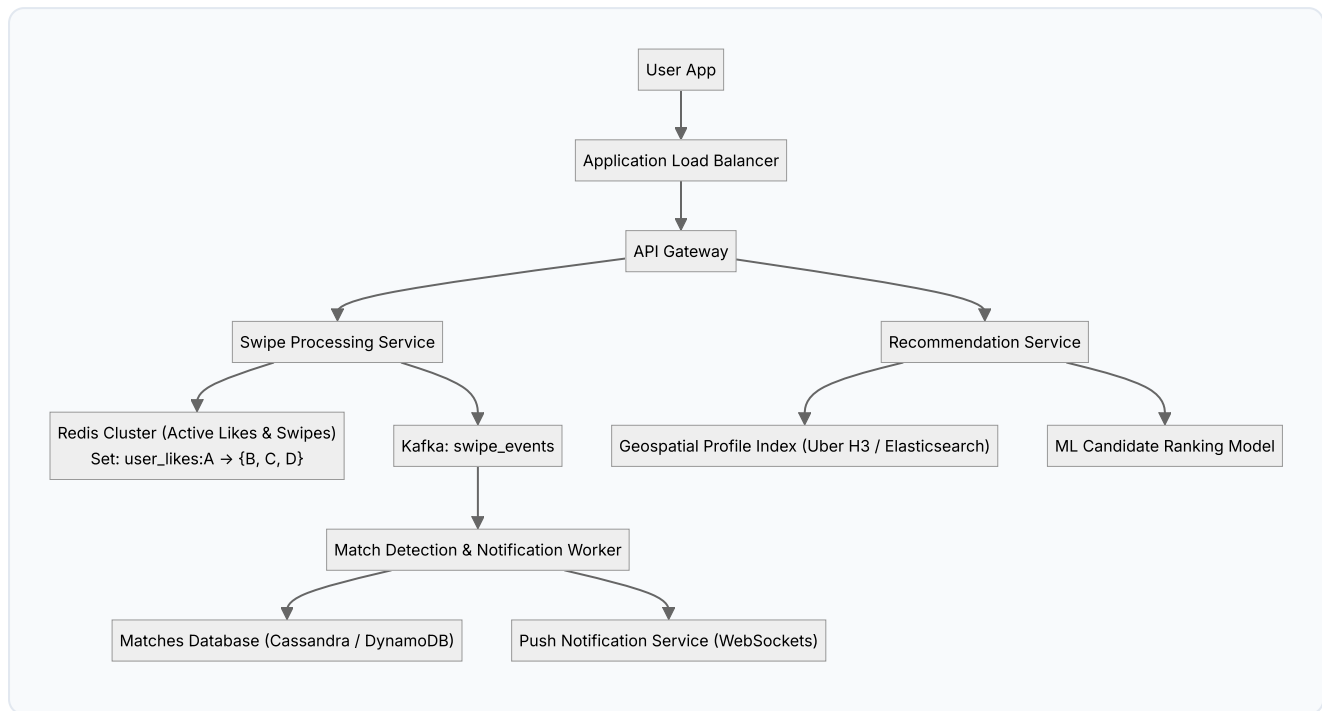
1. Problem Statement & Functional Requirements

Design the core discovery and mutual matching engine for a mobile dating application.

Requirements:

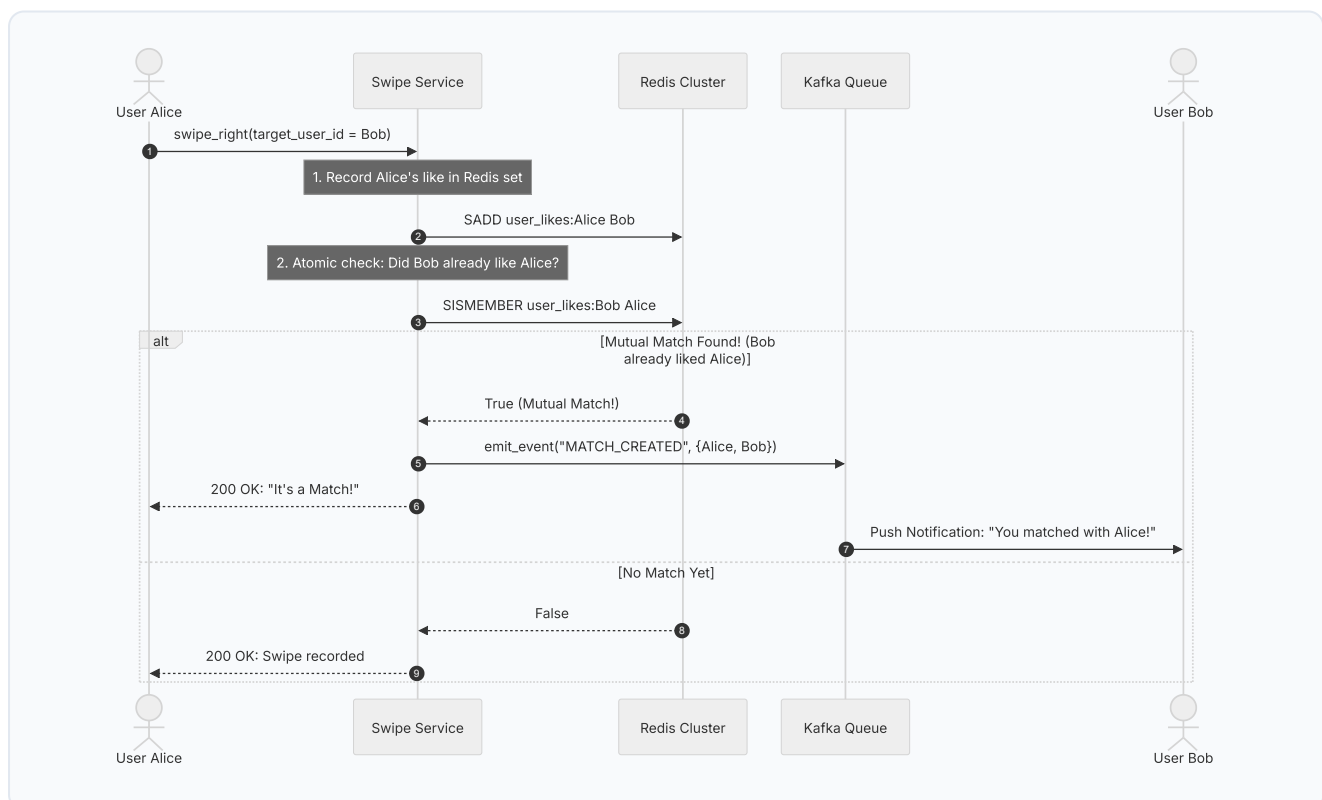
1. **Swipe Ingestion:** Users swipe Right (Like) or Left (Pass) at massive throughput (> 50,000 swipes/second).
 2. **Mutual Match Detection:** If both users swipe Right on each other, trigger a real-time **Mutual Match** event immediately.
 3. **Recommendation Feed:** Provide users with an endless stream of active, nearby profiles matching their preferences.
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2. High-Level Architecture Diagram



3. Real-Time Mutual Match Detection Pipeline

How do we check if a swipe produces a mutual match in under 10 milliseconds without querying heavy relational databases?



4. The 2-Stage Recommendation Engine

Delivering a fresh deck of 20 profiles to a user involves two distinct stages:

